

BACHELOR'S DEGREE (C.B.C.S) EXAMINATION, FEBRUARY / MARCH 2026
2024 ADMISSIONS REGULAR
SEMESTER IV - MAJOR (COMPUTER APPLICATIONS)
CA4DSEA01B24 - Advanced Machine Learning Concepts

Time : 1.5 Hours

Maximum Marks : 50

Part A

Answer all questions. Each question carries 1 mark.

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| 1. Differentiate between the General Boundary (G) and the Specific Boundary (S). | (10x1=10) |
| 2. Define a Target Function in the context of Machine Learning. | [CO1,Understand] |
| 3. State the primary goal of the Candidate Elimination Algorithm. | [CO1,Understand] |
| 4. Define an Artificial Neural Network (ANN). | [CO1,Understand] |
| 5. Differentiate between a Root Node and a Leaf Node. | [CO2,Remember] |
| 6. Explain the term Splitting in a Decision Tree. | [CO3,Remember] |
| 7. Describe the One Rule (1R) algorithm. | [CO3,Understand] |
| 8. Define Inductive Logic Programming (ILP). | [CO3,Understand] |
| 9. Define Abduction in the context of reasoning. | [CO4,Remember] |
| 10. Define a clause in the context of logic programming. | [CO4,Remember] |
| | [CO4,Understand] |

Part B

Answer any 2 questions from the bunch of CO1. Each question carries 5 marks.

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| 11. Explain the term Concept Class and differentiate it from a Concept. | (2x5=10) |
| 12. List the three steps in Input-Output Learning | [CO1,Understand] |
| 13. Describe the different ways to represent a Boolean Function. | [CO1,Understand] |
| | [CO1,Remember] |

Part B

Answer any 2 questions from the bunch of CO3. Each question carries 5 marks.

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| 14. Explain the Separate and Conquer strategy used in Rule Learning | (2x5=10) |
| 15. Explain Overfitting in Decision Trees. | [CO3,Understand] |
| 16. Explain the Divide and Conquer strategy used to build decision trees. | [CO3,Understand] |
| | [CO3,Understand] |

Part B

Answer any 2 questions from the bunch of CO4. Each question carries 5 marks.

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| 17. Explain the concept of Subsumption and Generalization in ILP. | (2x5=10) |
| 18. Explain the basic components of an ILP setting: Hypothesis (H), Background Knowledge (B), and Examples (E). | [CO4,Remember] |
| 19. Write similarities and dissimilarities between ILP and Decision Trees. | [CO4,Remember] |
| | [CO4,Understand] |

Part C

Answer any 1 question from the bunch of CO2. Each question carries 10 marks.

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| 20. Compare and Contrast the McCulloch-Pitts Neuron, Threshold Logic Unit (TLU), and Linear Machines. | (1x10=10) |
| | [CO2,Understand] |
| 21. Discuss the architecture and working of a Multi-Layer Perceptron (MLP) for solving the XOR problem. | [CO2,Understand] |